

Integrals – 6

Indefinite integrals: algorithms and non-elementarity

Homework

1 сентября 2026 года

Theoretical minimum. Partial fractions and Ostrogradsky's method (Hermite reduction); integration of $R(x, \sqrt{ax^2 + bx + c})$ and of $R(e^x)$; binomial differentials and Chebyshev's theorem; Liouville's theorem on elementary antiderivatives; the named non-elementary integrals erf, Si, li.

1. Apply Ostrogradsky's method to $\int \frac{x^2+1}{(x^2-1)^2(x+2)} dx$ and $\int \frac{dx}{(x^3-1)^2}$ without computing the complex roots of the denominator.
2. Compute $\int \frac{dx}{1-x+x^2-x^3}$ (Integration Bee 2016), $\int \frac{2x^3-1}{x(x^3+1)} dx$ (2019) and $\int \frac{x^{2024}-1}{x^{506}-1} dx$ (2024).
3. Compute $\int R(e^x) dx$ in general by the substitution $t = e^x$, and work out the examples $\int \frac{e^x}{e^x+2} dx$ (2018), $\int \frac{e^{x+1}}{e^x+1} dx$ (2025) and $\int \frac{dx}{1-e^{-x}}$ (2013).
4. Use Chebyshev's theorem to decide whether $\int \frac{dx}{\sqrt[3]{1+x^3}}$, $\int x^5(1+x^3)^{-1/2} dx$ and $\int \frac{\sqrt{1+\sqrt[4]{x}}}{\sqrt{x}} dx$ are elementary, and compute the elementary ones.
5. Prove that $\int \frac{dx}{\ln x}$ is not elementary, and express $\int \frac{dx}{\ln^2 x}$ and $\int \frac{x dx}{\ln x}$ through it.
6. (Integration Bee) Hyperbolic functions: $\int \tanh x dx$ (2016), $\int \operatorname{sech} x dx$ (2017), $\int \tanh^2 x dx$ (2018), $\int \frac{dx}{2+\cosh x}$ (2016) and $\int \sqrt{1+\cosh x} dx$ (2026).
7. (Integration Bee 2018) Compute $\int \frac{2017x^{2016}+2018x^{2017}}{1+x^{4034}+2x^{4035}+x^{4036}} dx$, noticing that the denominator is a perfect square.
8. Prove that the antiderivative of $\frac{1}{\sqrt{1-x^4}}$ is not elementary (the lemniscatic integral), and explain its place in the theory of elliptic integrals.

9. (Integration Bee) Trigonometry that reduces to a rational function:
 $\int \csc x \sec x \, dx$ (2018), $\int \frac{dx}{\tan x + \cot x}$ (2018), $\int \frac{\cos x + \cot x + \csc x + 1}{\sin x + \tan x + \sec x + 1} dx$ (2024)
and $\int \sqrt{\csc x - \sin x} \, dx$ (2015).
10. State the Hermite–Ostrogradsky algorithm as a procedure and explain in what sense it completes the problem of integrating rational functions: what exactly is left after it has been applied, and why the remainder is always a combination of logarithms and arctangents.